



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

cemc.uwaterloo.ca

**2023 Gauss Grade 7
Solutions**

1. Kiyana gives half of her 24 grapes to a friend. How many grapes does she give away?

- (A) 2 (B) 4 (C) 6 (D) 12 (E) 48

Source: 2023 Gauss Grade 7 #1

Primary Topics: Number Sense

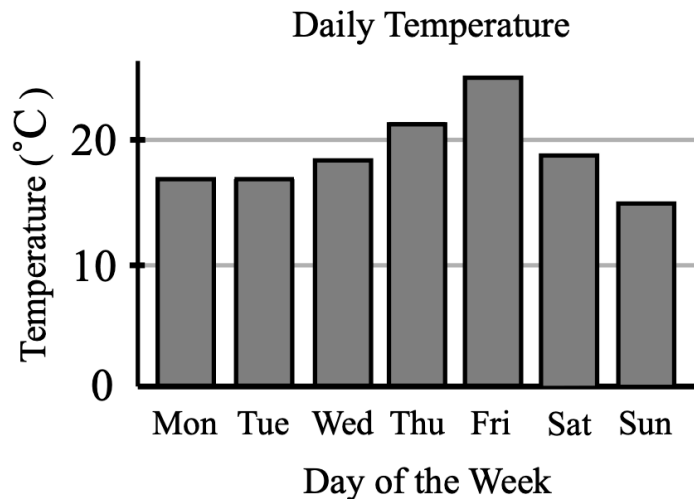
Secondary Topics: Fractions/Ratios | Expressions

Answer: D

Solution:

Half of 24 is $24 \div 2 = 12$. Kiyana gives her friend 12 grapes.

2. Based on the graph shown, which day of the week had the highest temperature?



- (A) Tuesday (B) Thursday (C) Friday (D) Saturday (E) Sunday

Source: 2023 Gauss Grade 7 #2

Primary Topics: Data Analysis

Secondary Topics: Counting | Graphs

Answer: C

Solution:

Reading from the graph, Friday had the highest temperature.

3. At a local farm, strawberries are sold at \$16.50 for each basket. What is the cost to buy 4 baskets of strawberries?

- (A) \$64.00 (B) \$66.00 (C) \$64.50 (D) \$65.50 (E) \$65.00

Source: 2023 Gauss Grade 7 #3

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Expressions | Fractions/Ratios | Rates | Counting

Answer: B

Solution:

At a cost of \$16.50 a basket, the cost to buy 4 baskets of strawberries is $4 \times \$16.50 = \66.00 .

4. The temperature last night was -5°C . It is now 3°C . How many degrees warmer is it now?

(A) 8°C (B) 3°C (C) 2°C (D) 13°C (E) 5°C

Source: 2023 Gauss Grade 7 #4

Primary Topics: Number Sense

Secondary Topics: Counting | Operations | Expressions

Answer: A

Solution:

The difference between 3 and -5 is $3 - (-5) = 3 + 5 = 8$. Therefore, it is now 8° warmer.

5. Sarah multiplied an integer by itself. Which of the following could be the result?

(A) 32 (B) 33 (C) 34 (D) 35 (E) 36

Source: 2023 Gauss Grade 7 #5

Primary Topics: Algebra and Equations | Number Sense

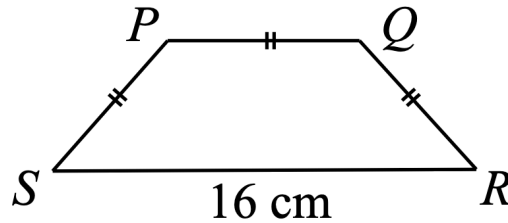
Secondary Topics: Operations | Exponents | Expressions

Answer: E

Solution:

Since $5 \times 5 = 25$ and each of the given answers is greater than 25, then the integer that Sarah multiplied by itself must have been greater than 5. Further, $6 \times 6 = 36$ and each of the given answers is less than or equal to 36. Thus, of the given answers, only 36 could be the result of multiplying an integer by itself. Alternatively, we may have noted that the result of multiplying an integer by itself is a perfect square, and of the answers given, 36 is the only perfect square.

6. In the figure shown, $PQRS$ has three sides of equal length and $SR = 16$ cm.



If the perimeter of $PQRS$ is 40 cm, then the length of PQ is

- (A) 6 cm (B) 7 cm (C) 8 cm (D) 9 cm (E) 10 cm

Source: 2023 Gauss Grade 7 #6

Primary Topics: Geometry and Measurement

Secondary Topics: Operations | Quadrilaterals | Perimeter | Polygons

Answer: C

Solution:

Since the perimeter of $PQRS$ is 40 cm and $SR = 16$ cm, then the combined length of the remaining three sides is $40 \text{ cm} - 16 \text{ cm} = 24 \text{ cm}$. Each of the remaining three sides is equal in length, and so $PQ = \frac{24 \text{ cm}}{3} = 8 \text{ cm}$.

7. Which of the following is equal to a whole number?

- (A) $\frac{52}{5}$ (B) $\frac{52}{7}$ (C) $\frac{52}{4}$ (D) $\frac{52}{3}$ (E) $\frac{52}{6}$

Source: 2023 Gauss Grade 7 #7

Primary Topics: Number Sense

Secondary Topics: Fractions/Ratios | Operations | Expressions

Answer: C

Solution:

Dividing 52 by each of the given denominators, we get that $\frac{52}{4} = 13$ is the only whole number.

8. A circle has a radius of 4 cm. A line segment joins two points on the circle. What is the greatest possible length of the line segment?

- (A) 10 cm (B) 8 cm (C) 4 cm (D) 12 cm (E) 6 cm

Source: 2023 Gauss Grade 7 #8

Primary Topics: Geometry and Measurement

Secondary Topics: Circles | Measurement

Answer: B

Solution:

The line segment with greatest length that joins two points on a circle is a diameter of the circle. Since the circle has a radius of 4 cm, then its diameter has length $2 \times 4 \text{ cm} = 8 \text{ cm}$, and so the greatest possible length of the line segment is 8 cm.

9. An integer is randomly chosen from the list 10, 11, 12, 13, 14, 15, 16, 17, 18, 19.

What is the probability that the chosen integer is even?

- (A) $\frac{3}{10}$ (B) $\frac{4}{10}$ (C) $\frac{5}{10}$ (D) $\frac{6}{10}$ (E) $\frac{7}{10}$

Source: 2023 Gauss Grade 7 #9

Primary Topics: Counting and Probability

Secondary Topics: Counting | Probability

Answer: C

Solution:

The number of integers in the list is 10. Of these integers, 5 are even (they are 10, 12, 14, 16, and 18). Thus, the probability that the chosen integer is even is $\frac{5}{10}$.

10. The grocery receipt shows the cost of three items before tax is added.

Sponge	\$4.20
Shampoo	\$7.60
Soap	\$3.20

When a 5% tax is added to the cost of the items, what is the total cost for the three items?

- (A) \$15.16 (B) \$15.08 (C) \$15.22 (D) \$15.75 (E) \$15.38

Source: 2023 Gauss Grade 7 #10

Primary Topics: Algebra and Equations | Other

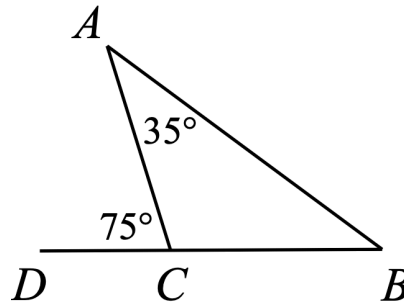
Secondary Topics: Decimals | Percentages | Expressions

Answer: D

Solution:

Before adding tax, the combined cost of the three items is $\$4.20 + \$7.60 + \$3.20 = \15.00 . The 5% tax on \$15.00 is $0.05 \times \$15.00 = \0.75 , and so the total cost of the three items, after tax is added, is $\$15.00 + \$0.75 = \$15.75$. Note that we could have calculated the 5% tax on each individual item, however doing so is less efficient than calculating tax on the \$15.00 total.

11. In the diagram, BCD is a straight line segment.



The measure of $\angle ABC$ is

- (A) 35° (B) 40° (C) 60° (D) 75° (E) 45°

Source: 2023 Gauss Grade 7 #11

Primary Topics: Geometry and Measurement

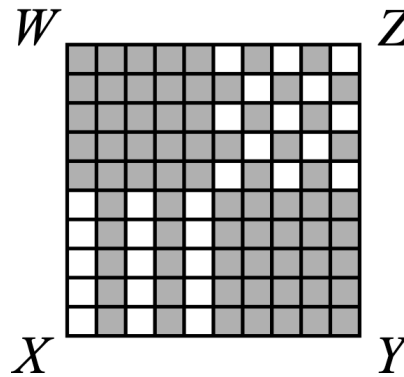
Secondary Topics: Triangles | Angles | Equations Solving | Counting

Answer: B

Solution:

Since BCD is a straight line segment, then $\angle BCD = 180^\circ$. Therefore, $\angle ACB = \angle BCD - \angle ACD = 180^\circ - 75^\circ = 105^\circ$. Since the sum of the three angles in $\triangle ABC$ is 180° , then $\angle ABC = 180^\circ - 105^\circ - 35^\circ = 40^\circ$.

12. Square $WXYZ$ is divided into 100 small identical squares. Some small squares are shaded and some are unshaded, as shown.



How many more of the small squares need to be shaded so that 75% of the area of $WXYZ$ is shaded?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Source: 2023 Gauss Grade 7 #12

Primary Topics: Number Sense | Geometry and Measurement

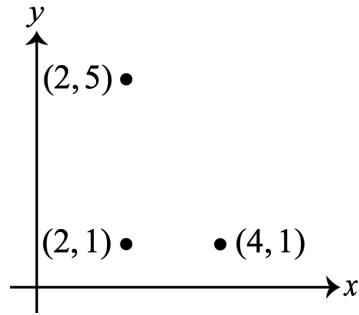
Secondary Topics: Percentages | Counting | Area | Quadrilaterals

Answer: A

Solution:

Of the 100 small identical squares, 28 are presently unshaded, and so $100 - 28 = 72$ are shaded. So that 75% of the area of $WXYZ$ is shaded, 75 of the 100 small squares must be shaded. Therefore, $75 - 72 = 3$ more of the small squares must be shaded.

13. In the diagram, the points $(2, 1)$, $(4, 1)$ and $(2, 5)$ are three vertices of a rectangle.



What are the coordinates of the fourth vertex of the rectangle?

- (A) $(5, 2)$ (B) $(4, 4)$ (C) $(1, 5)$ (D) $(4, 5)$ (E) $(2, 4)$

Source: 2023 Gauss Grade 7 #13

Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Quadrilaterals | Counting | Polygons

Answer: D

Solution:

Suppose we call the unknown vertex V . The side of the rectangle joining the points $(2, 1)$ and $(2, 5)$ is vertical, and so the opposite side of the rectangle (the side joining $(4, 1)$ to V) must also be vertical. This means that V has the same x -coordinate as $(4, 1)$, which is 4. Similarly, the side of the rectangle joining the points $(2, 1)$ and $(4, 1)$ is horizontal, and so the opposite side of the rectangle (the side joining $(2, 5)$ to V) must also be horizontal. This means that V has the same y -coordinate as $(2, 5)$, which is 5. Therefore, the coordinates of the fourth vertex of the rectangle are $(4, 5)$.

14. The sum of two different prime numbers is 10. The product of these two numbers is

- (A) 24 (B) 16 (C) 4 (D) 21 (E) 9

Source: 2023 Gauss Grade 7 #14

Primary Topics: Number Sense

Secondary Topics: Operations | Factoring | Prime Numbers

Answer: D

Solution:

The prime numbers that are less than 10 are 2, 3, 5, and 7. Thus, the only two different prime numbers whose sum is 10 are 3 and 7. The product of these two numbers is $3 \times 7 = 21$.

15. Suppose n is a number such that the mean (average) of the list of numbers 2, 9, 4, n , $2n$ is equal to 6. What is the value of n ?
- (A) 9 (B) 12 (C) 10 (D) 5 (E) 6

Source: 2023 Gauss Grade 7 #15

Primary Topics: Number Sense | Data Analysis

Secondary Topics: Averages | Expressions

Answer: D

Solution:

The given list, 2, 9, 4, n , $2n$ contains 5 numbers. The average of these 5 numbers is 6, and so the sum of the 5 numbers is $5 \times 6 = 30$. That is, $2 + 9 + 4 + n + 2n = 30$ or $15 + 3n = 30$, and so $3n = 15$ or $n = 5$.

16. Each number from 1 to 6 replaces one of the letters P , Q , R , S , T , and U . The sum of P and Q is 5 and the difference between R and S is 5. If T is greater than U , what number replaces the letter T ?
- (A) 4 (B) 6 (C) 2 (D) 3 (E) 5

Source: 2023 Gauss Grade 7 #16

Primary Topics: Number Sense | Algebra and Equations

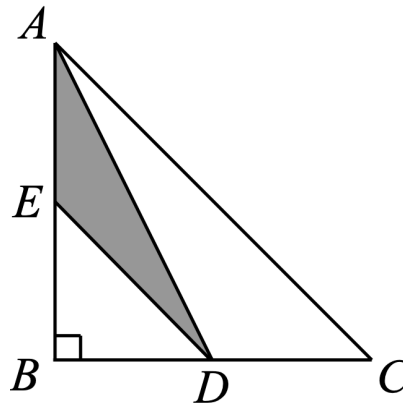
Secondary Topics: Patterning/Sequences/Series | Expressions | Inequalities

Answer: E

Solution:

The sum of P and Q is equal to 5, and so P and Q are (in some order) either equal to 1 and 4, or they are equal to 2 and 3. The difference between R and S is equal to 5, and so R and S must be (in some order) equal to 1 and 6. Since R and S are equal to 1 and 6, then neither P nor Q can be 1, which means that P and Q cannot be equal to 1 and 4, and so they must be equal to 2 and 3. The only numbers from 1 to 6 not accounted for are 4 and 5. Since T is greater than U , then the number that replaces T is 5.

17. In the diagram, $\triangle ABC$ is a right-angled isosceles triangle. D is the midpoint of BC and E is the midpoint of AB .



If $AB = BC = 24$ cm, what is the area of $\triangle AED$?

- (A) 48 cm^2 (B) 36 cm^2 (C) 72 cm^2 (D) 9 cm^2 (E) 54 cm^2

Source: 2023 Gauss Grade 7 #17

Primary Topics: Geometry and Measurement

Secondary Topics: Polygons | Triangles | Area

Answer: C

Solution:

Solution 1

The area of $\triangle AED$ is equal to one-half its base times its height. Suppose the base of $\triangle AED$ is AE , then its height is BD (AE is perpendicular to BD). Since $AB = BC = 24$ cm and E and D are the midpoints of their respective sides, then $AE = 12$ cm and $BD = 12$ cm. Thus, the area of $\triangle AED$ is $\frac{1}{2} \times 12 \text{ cm} \times 12 \text{ cm} = 72 \text{ cm}^2$.

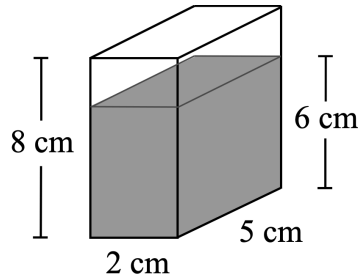
Solution 2

The area of $\triangle AED$ is equal to the area of $\triangle ABD$ minus the area of $\triangle EBD$. Suppose the base of $\triangle EBD$ is BD , then its height is EB . Since $AB = BC = 24$ cm and E and D are the midpoints of their respective sides, then $EB = 12$ cm and $BD = 12$ cm. Thus, the area of $\triangle EBD$ is $\frac{1}{2} \times 12 \text{ cm} \times 12 \text{ cm} = 72 \text{ cm}^2$.

The area of $\triangle ABD$ is equal to $\frac{1}{2} \times BD \times AB = \frac{1}{2} \times 12 \text{ cm} \times 24 \text{ cm} = 144 \text{ cm}^2$.

Thus, the area of $\triangle AED$ is $144 \text{ cm}^2 - 72 \text{ cm}^2 = 72 \text{ cm}^2$.

18. A closed rectangular prism with height 8 cm is standing on a face with dimensions 2 cm by 5 cm. The prism contains water with a depth of 6 cm, as shown.



When the prism is tipped so that it stands on a face with the greatest area, the depth of the water is

- (A) 0.75 cm (B) 1 cm (C) 1.25 cm (D) 1.5 cm (E) 1.75 cm

Source: 2023 Gauss Grade 7 #18

Primary Topics: Geometry and Measurement

Secondary Topics: Volume | Prisms | Measurement

Answer: D

Solution:

The water is in the shape of a rectangular prism with a 2 cm by 5 cm base and depth 6 cm. Therefore, the volume of water is $2 \text{ cm} \times 5 \text{ cm} \times 6 \text{ cm} = 60 \text{ cm}^3$. A face of the prism having the greatest area has dimensions 5 cm by 8 cm. When the prism is tipped so that it stands on a 5 cm by 8 cm face, the water is once again in the shape of a rectangular prism with a 5 cm by 8 cm base and unknown depth. Suppose that after the prism is tipped, the water's depth is d cm. Since the volume of water is still 60 cm^3 when the prism is tipped, then $5 \text{ cm} \times 8 \text{ cm} \times d \text{ cm} = 60 \text{ cm}^3$ or $40d \text{ cm}^3 = 60 \text{ cm}^3$, and so $d = \frac{60}{40} = \frac{3}{2}$. When the prism is tipped so that it stands on a face with the greatest area, the depth of the water is $\frac{3}{2} \text{ cm} = 1.5 \text{ cm}$.

19. Two standard dice are rolled. The product of the two numbers rolled is calculated.

What is the probability that the ones digit of this product is 0?

- (A) $\frac{11}{36}$ (B) $\frac{2}{9}$ (C) $\frac{1}{36}$ (D) $\frac{1}{6}$ (E) $\frac{5}{36}$

Source: 2023 Gauss Grade 7 #19

Primary Topics: Counting and Probability

Secondary Topics: Games | Counting | Probability | Fractions/Ratios

Answer: D

Solution:

Solution 1

We begin by completing a table in which the ones digit of each possible product is listed. For example, when the number on the first die is 3 and the number on the second die is 6, the entry in the table is 8 since $3 \times 6 = 18$ and the ones digit of 18 is 8.

Of the 36 possible outcomes in the table above, 6 outcomes have a ones digit that is equal to 0. Thus, the probability that the ones digit of the product is 0 is $\frac{6}{36} = \frac{1}{6}$.

Solution 2

Since the ones digit of the product is 0, then the product is divisible by 5 and is even. Since the possible numbers in the product are 1, 2, 3, 4, 5, 6, then one of the numbers rolled must be 5. Since the product is even (and 5 is not), then the other number rolled must be one of the three even numbers, namely 2, 4, 6. Thus, the possible pairs of numbers that can be rolled to give a product whose ones digit is 0, are (5, 2), (5, 4), (5, 6) or (2, 5), (4, 5), (6, 5). (We note that the first number in the ordered pair represents the first number rolled, while the second number in the pair is the second number rolled.) Since there are 6 possible rolls for each of the two dice, then there are $6 \times 6 = 36$ possible ordered pairs representing all possible outcomes. Since 6 of these ordered pairs represent a product whose ones digit is 0, then the required probability is $\frac{6}{36} = \frac{1}{6}$.

20. How many pairs of positive integers a and b satisfy the equation $\frac{a}{7} + \frac{2}{b} = 1$?

(A) 4

(B) 1

(C) 0

(D) 5

(E) 2

Source: 2023 Gauss Grade 7 #20

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Equations Solving | Fractions/Ratios | Operations

Answer: E

Solution:

Since a and b are positive integers, then each of $\frac{a}{7}$ and $\frac{2}{b}$ is greater than 0. The sum of $\frac{a}{7}$ and $\frac{2}{b}$ is equal to 1, and so each is less than 1. Since $\frac{a}{7}$ is greater than 0 and less than 1, then the possible values of a are 1, 2, 3, 4, 5, 6. By substituting each of these values for a into the equation one at a time, we can determine if there is a positive integer value of b for which the equation is true.

Substituting $a = 1$, we get $\frac{1}{7} + \frac{2}{b} = 1$ or $\frac{2}{b} = 1 - \frac{1}{7}$, and so $\frac{2}{b} = \frac{6}{7}$.

Since $\frac{2}{b} = \frac{6}{7}$, we can multiply the numerator and denominator of the first fraction by 3 (which is $6 \div 2$) to get $\frac{6}{3b} = \frac{6}{7}$.

This gives $3b = 7$ which does not have an integer solution ($b = \frac{7}{3}$). Thus when $a = 1$, there is no positive integer value of b that satisfies the equation.

Substituting $a = 2$, we get $\frac{2}{7} + \frac{2}{b} = 1$ or $\frac{2}{b} = 1 - \frac{2}{7}$, and so $\frac{2}{b} = \frac{5}{7}$. Since $\frac{2}{b} = \frac{5}{7}$, we can multiply the numerator and denominator of the first fraction by 5, and the numerator and denominator of the second fraction by 2 to get $\frac{10}{5b} = \frac{10}{14}$. This gives $5b = 14$ which does not have an integer solution.

Thus when $a = 2$, there is no positive integer value of b that satisfies the equation.

Substituting $a = 3$, we get $\frac{3}{7} + \frac{2}{b} = 1$ or $\frac{2}{b} = 1 - \frac{3}{7}$, and so $\frac{2}{b} = \frac{4}{7}$. Since $\frac{2}{b} = \frac{4}{7}$, we can multiply the numerator and denominator of the first fraction by 2 to get $\frac{4}{2b} = \frac{4}{7}$. This gives $2b = 7$ which does not have an integer solution. Thus when $a = 3$, there is no positive integer value of b that satisfies the equation.

Substituting $a = 4$ and simplifying, we get $\frac{2}{b} = \frac{3}{7}$. Since $\frac{2}{b} = \frac{3}{7}$, we can multiply the numerator and denominator of the first fraction by 3, and the numerator and denominator of the second fraction by 2 to get $\frac{6}{3b} = \frac{6}{14}$.

This gives $3b = 14$ which does not have an integer solution.

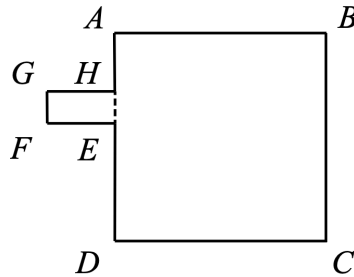
Thus when $a = 4$, there is no positive integer value of b that satisfies the equation.

Substituting $a = 5$ and simplifying, we get $\frac{2}{b} = \frac{2}{7}$. Since the numerators are equal, then the denominators must be equal, and so $b = 7$ satisfies the equation.

Finally, substituting $a = 6$ and simplifying, we get $\frac{2}{b} = \frac{1}{7}$. Since $\frac{2}{b} = \frac{1}{7}$, we can multiply the numerator and denominator of the second fraction by 2 to get $\frac{2}{b} = \frac{2}{14}$, and so $b = 14$.

Thus, there are two pairs of positive integers a and b that satisfy the given equation: $a = 5, b = 7$ and $a = 6, b = 14$.

21. Eight-sided polygon $ABCDEFGH$ has integer side lengths. It can be divided into a rectangle and a square, as shown.



The area of the square is greater than the area of the rectangle. The product of the two areas is equal to 98. Which of the following could be the perimeter of $ABCDEFGH$?

- (A) 51 (B) 32 (C) 44 (D) 34 (E) 33

Source: 2023 Gauss Grade 7 #21

Primary Topics: Geometry and Measurement

Secondary Topics: Area | Perimeter | Measurement | Logic

Answer: B

Solution:

Since $ABCD$ is a square and its side lengths are integers, then its area is equal to a perfect square. Since the product of the areas of $ABCD$ and $EFGH$ (the rectangle) is equal to 98, then the area of $ABCD$ is a divisor of 98. The positive divisors of 98 are 1, 2, 7, 14, 49, and 98. There are exactly two divisors of 98 that are perfect squares, namely 1 and 49. Since the area of $ABCD$ is greater than the area of $EFGH$, then the area of $ABCD$ is 49, and so the area of $EFGH$ is 2 (since $49 \times 2 = 98$). Square $ABCD$ has area 49, and so $AB = BC = CD = DA = 7$.

The perimeter of $ABCDEFGH$ is equal to

$$\begin{aligned}
 & AB + BC + CD + DE + EF + FG + GH + HA \\
 &= 7 + 7 + 7 + DE + EF + EH + GH + HA && \text{(since } EH = FG\text{)} \\
 &= 21 + DE + EH + HA + EF + GH && \text{(reorganizing)} \\
 &= 21 + DA + EF + GH && \text{(since } DE + EH + HA = DA\text{)} \\
 &= 21 + 7 + EF + GH && \text{(since } DA = 7\text{)} \\
 &= 28 + 2 \times GH && \text{(since } EF = GH\text{)}
 \end{aligned}$$

Since the side lengths are integers and the area of $EFGH$ is 2, then either $GH = 1$ (and $FG = 2$), or $GH = 2$ (and $FG = 1$). If $GH = 1$, then the perimeter of $ABCDEFGH$ is $28 + 2 \times 1 = 30$. Since 30 is not given as a possible answer, then $GH = 2$ and the perimeter is $28 + 2 \times 2 = 32$.

22. A *Gareth sequence* is a sequence of numbers in which each number after the second is the *non-negative* difference between the two previous numbers. For example, if a Gareth sequence begins 15, 12, then

- the third number in the sequence is $15 - 12 = 3$,
- the fourth number is $12 - 3 = 9$,
- the fifth number is $9 - 3 = 6$,

and so the resulting sequence is 15, 12, 3, 9, 6, ... If a Gareth sequence begins 10, 8, what is the sum of the first 30 numbers in the sequence?

- (A) 40 (B) 72 (C) 34 (D) 56 (E) 64

Source: 2023 Gauss Grade 7 #22

Primary Topics: Data Analysis

Secondary Topics: Patterning/Sequences/Series | Counting | Operations

Answer: E

Solution:

If a Gareth sequence begins 10, 8, then the 3rd number in the sequence is $10 - 8 = 2$, the 4th is $8 - 2 = 6$, the 5th is $6 - 2 = 4$, the 6th is $6 - 4 = 2$, the 7th is $4 - 2 = 2$, the 8th is $2 - 2 = 0$, the 9th is $2 - 0 = 2$, the 10th is $2 - 0 = 2$, and the 11th is $2 - 2 = 0$. Thus, the resulting sequence is 10, 8, 2, 6, 4, 2, 2, 0, 2, 2, 0, ... The first 5 numbers in the sequence are 10, 8, 2, 6, 4, the next 3 numbers are 2, 2, 0, and this block of 3 numbers appears to repeat. Since each new number added to the end of this sequence is determined by the two previous numbers in the sequence, then this block of 3 numbers will indeed continue to repeat. (That is, since the block repeats once, then it will continue repeating.) The first 30 numbers of the sequence begins with the first 5 numbers, followed by 8 blocks of 2, 2, 0, followed by one additional 2 (since $5 + 8 \times 3 + 1 = 30$). The sum of the first 5 numbers is $10 + 8 + 2 + 6 + 4 = 30$. The sum of each repeating block is $2 + 2 + 0 = 4$, and so the sum of 8 such blocks is $8 \times 4 = 32$. Thus, the sum of the first 30 numbers in the sequence is $30 + 32 + 2 = 64$.

23. The digits from 1 to 9 are each used exactly once to write three one-digit integers and three two-digit integers. The one-digit integers are equal to the length, width and height of a rectangular prism. The two-digit integers are equal to the areas of the faces of the same prism. What is the surface area of the rectangular prism?
- (A) 176 (B) 184 (C) 186 (D) 198 (E) 212

Source: 2023 Gauss Grade 7 #23

Primary Topics: Number Sense | Geometry and Measurement

Secondary Topics: Digits | Area | Prisms | Surface Area

Answer: D

Solution:

Suppose that the length, or the width, or the height of the rectangular prism is equal to 5. The product of 5 with any of the remaining digits has a units (ones) digit that is equal to 5 or it is equal to 0. This means that if the length, or the width, or the height of the rectangular prism is equal to 5, then at least one of the two-digit integers (the area of a face) has a units digit that is equal to 5 or 0.

However, 0 is not a digit that can be used, and each digit from 1 to 9 is used exactly once (that is, 5 cannot be used twice), and so it is not possible for one of the dimensions of the rectangular prism to equal 5. Thus, the digit 5 occurs in one of the two-digit integers (the area of a face). The digit 5 cannot be the units digit of the area of a face, since this would require that one of the dimensions be 5. Therefore, one of the areas of a face has a tens digit that is equal to 5. The two-digit integers with tens digit 5 that are equal to the product of two different one-digit integers (not equal to 5) are $54 = 6 \times 9$ and $56 = 7 \times 8$. Suppose that two of the dimensions of the prism are 7 and 8, and so one of the areas is 56. In this case, the digits 5, 6, 7, and 8 have been used, and so the digits 1, 2, 3, 4, and 9 remain. Which of these digits is equal to the remaining dimension of the prism? It cannot be 1 since the product of 1 and 7 does not give a two-digit area, nor does the product of 1 and 8. It cannot be 2 since the product of 2 and 8 is 16 and the digit 6 has already been used. It cannot be 3 since $3 \times 7 = 21$ and $3 \times 8 = 24$, and so the areas of two faces share the digit 2. It cannot be 4 since $4 \times 7 = 28$ and the digit 8 has already been used. Finally, it cannot be 9 since $9 \times 7 = 63$ and the digit 6 has already been used. Therefore, it is not possible for 7 and 8 to be the dimensions of the prism, and thus 6 and 9 must be two of the three dimensions. Using a similar systematic check of the remaining digits, we determine that 3 is the third dimension of the prism. That is, when the dimensions of the prism are 3, 6 and 9, the areas of the faces are $3 \times 6 = 18$, $3 \times 9 = 27$, and $6 \times 9 = 54$, and we may confirm that each of the digits from 1 to 9 has been used exactly once. Since the areas of the faces are 18, 27 and 54, the surface area of the rectangular prism is $2 \times (18 + 27 + 54)$ or $2 \times 99 = 198$.

24. A circle is divided into six equal sections. Each section is to be coloured with a single colour so that three sections are red, one is blue, one is green, and one is yellow. Two circles have the same colouring if one can be rotated to match the other. In the diagram, Figure 1 and Figure 2 have the same colouring, while Figure 1 and Figure 3 have different colourings.

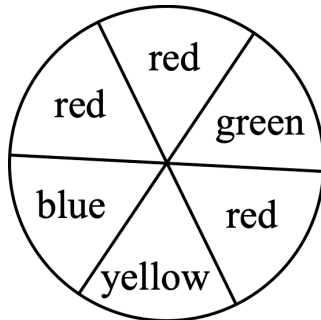


Figure 1

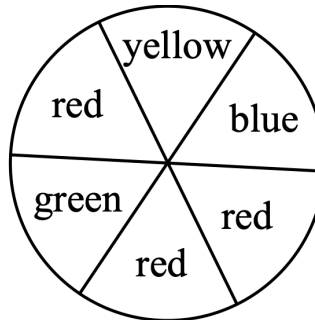


Figure 2

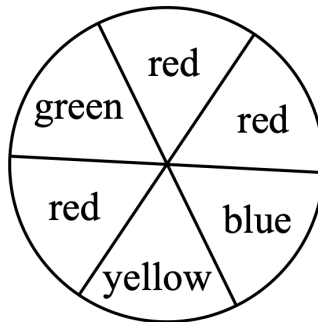


Figure 3

How many different colourings are there for the circle?

- (A) 14 (B) 12 (C) 24 (D) 10 (E) 20

Source: 2023 Gauss Grade 7 #24

Primary Topics: Counting and Probability

Secondary Topics: Counting | Circles

Answer: E

Solution:

Solution 1

Begin by colouring the section at the top blue. Since two circles have the same colouring if one can be rotated to match the other, it does not matter which section is coloured blue, so we arbitrarily choose the top section.

There are now 5 sections which can be coloured green. After choosing the section to be coloured green, there are 4 sections remaining which can be coloured yellow. Each of the remaining 3 sections must then be coloured red. Thus, the total number of different colourings of the circle is $5 \times 4 = 20$.

Solution 2

We begin by considering the locations of the three sections coloured red, relative to one another. The three red sections could be adjacent to one another, or exactly two red sections could be adjacent to one another, or no red section could be adjacent to another red section. We consider each of these 3 cases separately.

Case 1: All three red sections are adjacent to one another.

Begin by colouring any three adjacent sections red.

Since two circles have the same colouring if one can be rotated to match the other, it does not matter which three adjacent sections are coloured red. Consider the first section that follows the three red sections as we move clockwise around the circle. There are 3 choices for the colour of this section: blue, green or yellow. Continuing to move clockwise to the next section, there are now 2 choices for the colour of this section. Finally, there is 1 choice for the colour of the final section, and thus there are $3 \times 2 \times 1 = 6$ different colourings of the circle in which all three red sections are adjacent to one another. These 6 colourings are shown below.

Case 2: Exactly two red sections are adjacent to one another.

There are two different possible arrangements in which exactly two red sections are adjacent to one another. In the first of these, the next two sections that follow the two adjacent red sections as we move clockwise around the circle, are both not red. We call this Case 2a. In the second of these, the section that follows the two adjacent red sections as we move clockwise around the circle is not red, but the next section is. We call this Case 2b. The arrangements for Cases 2a and 2b are shown below. Notice that the first of these two circles cannot be rotated to match the second. The number of colourings in Case 2a and in Case 2b are each equal to the number of colourings as in Case 1. That is, there are 3 choices for the first uncoloured section that follows the two red sections as we move clockwise around the circle. Continuing to move clockwise to the next uncoloured section, there are now 2 choices for the colour of this section. Finally, there is 1 choice for the colour of the final section, and thus there are $3 \times 2 \times 1 = 6$ different colourings of the circle in Case 2a as well as in Case 2b. These 12 colourings are shown below.

Case 3: No red section is adjacent to another red section.

Begin by colouring any three non-adjacent sections red.

Since two circles have the same colouring if one can be rotated to match the other, it does not matter which three non-adjacent sections are coloured red. In this case, there are 2 possible colourings as shown below.

A circle with any other arrangement of the green, yellow and blue sections can be rotated to match one of the two circles above.

The total number of different colourings of the circle is $6 + 12 + 2 = 20$.

25. A school trip offered its participants three activities: hiking, canoeing and swimming. Attendance records show that of all participants
- 10 students participated in all three activities,
 - 50% participated in at least hiking and canoeing,
 - 60% participated in at least hiking and swimming,
 - $k\%$ participated in at least canoeing and swimming, and
 - no students participated in fewer than two activities.

If k is a positive integer, what is the sum of all possible values of k ?

- (A) 191 (B) 185 (C) 261 (D) 95 (E) 175

Source: 2023 Gauss Grade 7 #25

Primary Topics: Counting and Probability

Secondary Topics: Counting | Graphs | Logic

Answer: B

Solution:

We can represent the given information in a Venn diagram by first introducing some variables. Let x be the number of students that participated in hiking and canoeing, but not swimming. Let y be the number of students that participated in hiking and swimming, but not canoeing. Let z be the number of students that participated in canoeing and swimming, but not hiking. Since 10 students participated in all three activities and no students participated in fewer than two activities, we complete the Venn diagram as shown.

Suppose that the total number of students participating in the school trip was n . Since 50% of all students participated in at least hiking and canoeing, then $\frac{50}{100}n$ or $\frac{n}{2}$ participated in at least hiking and canoeing. Since this number of students, $\frac{n}{2}$, is an integer, then n must be divisible by 2.

Similarly, $\frac{60}{100}n$ or $\frac{3n}{5}$ students participated in at least hiking and swimming. Since this number of students, $\frac{3n}{5}$, is an integer, then n must be divisible by 5 (since 3 and 5 have no factors in common). This means that n is divisible by both 2 and 5, and thus n is divisible by 10. From the Venn diagram, we see that $x + 10 = \frac{n}{2}$, and $y + 10 = \frac{3n}{5}$. Since the total number of participants is n , we also get that $x + y + z + 10 = n$ or $z = n - 10 - x - y$. We may now use these equations,

$$x = \frac{n}{2} - 10, y = \frac{3n}{5} - 10, \text{ and } z = n - 10 - x - y$$

and the fact that n is divisible by 10, to determine all possible values of z . We can then use the values of z to determine all possible values of the positive integer k , where $k\%$ participated in at least canoeing and swimming.

Since n is a positive integer that is divisible by 10, its smallest possible value is 10. However, substituting $n = 10$ into $x = \frac{n}{2} - 10$, we get $x = 5 - 10$ and so $x = -5$ which is not possible. (Recall that x is the number of students that participated in hiking and canoeing, but not swimming, and so $x \geq 0$.) Next, we try $n = 20$. When $n = 20$, $x = 10 - 10$ and so $x = 0$.

When $n = 20$, $y = \frac{3 \times 20}{5} - 10$ or $y = 12 - 10$, and so $y = 2$.

Finally, when $n = 20$, $x = 0$, and $y = 2$, we get $z = 20 - 10 - 0 - 2 = 8$. When $z = 8$, the number of students who participated in at least canoeing and swimming is $8 + 10 = 18$ (since 10 students participated in all three), and so the percentage of students who participated in at least canoeing and swimming is $\frac{18}{20} \times 100\% = 90\%$, and so $k = 90$. In the table below, we continue in this way by using successively greater multiples of 10 for the value of n .

For values of n that are greater than 100, we get that $z < 0$, which is not possible. Therefore, the sum of all such positive integers k is $90 + 40 + 30 + 15 + 10 = 185$.